Principles of Quantum Mechanics – R. Shankar

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1 Mathematical Introduction (pp. 1 – 73)

1.1 The Delta function

In discrete space with basis $|x_i\rangle$,

- Orthogonality: $\langle x_i|x_j\rangle = \delta_{ij}$
- Completeness: $\sum_{i=1}^{n} |x_i\rangle \langle x_i| = I$

Natural generalization to continuous functions.

- Orthogonality: $\langle x|x'\rangle = \delta(x - x')$
- Completeness: $\int_a^b |x'\rangle \langle x| \, dx = I$

δ is the Dirac delta function, defined as

$$\boxed{\begin{gathered} \delta(x - x') = 0 \ , \quad x \neq x' \\ \int_a^b \delta(x - x') \, dx' = 1 \ , \quad a < x < b \end{gathered}}$$

Sometimes we write $\delta(0) = \infty$, but this isn't well defined except in the context of an integral. The δ also works like a "sampling function":

$$\int \delta(x - x')f(x')\ dx' = f(x)$$

Derivative: The derivative of the delta function acts like an operator

$$\delta'(x - x') = \delta(x - x')\frac{d}{dx}$$

To see this, note that

$$\int \delta'(x - x')f(x')\ dx' = \frac{d}{dx}\int \delta(x - x')f(x')\ dx' = \frac{df(x)}{dx}$$

Fourier Series: We can also represent the delta function using a fourier series. Recall that

$$\text{Fourier transform:}\quad f(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ikx}f(x)\ dx$$
$$\text{Inverse transform:}\quad f(x') = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{ikx'}f(k)\ dk$$

$$f(x') = \int_{-\infty}^{\infty}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(x'-x)}\ dk\right)f(x)\ dx$$

Reasoning from the use of delta as a sampling function,

$$\delta(x' - x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(x'-x)}\ dk$$

1.2 The Derivative Operator

We can have operators in infinite dimensions, e.g.

$$\Omega\,|f\rangle = |\tilde{f}\rangle$$

The derivative is one such operator:

$$D\,|f\rangle = |\frac{df}{dx}\rangle$$

The elements of D in the x basis are

$$D_{xx'} \equiv \langle x|D|x'\rangle = \delta'(x - x') = \delta(x - x')\frac{d}{dx'}$$

To see this, recall the use of δ as a sampling function.

$$\langle x|D|f\rangle = \langle x|\frac{df}{dx}\rangle = \frac{df(x)}{dx}$$
$$\int \langle x|D|x'\rangle\,\langle x'|f\rangle\,dx' = \frac{df}{dx}$$

1.2.1 The Hermitian Operator $K = -iD$

However, D is not Hermitian: $D_{xx'} \neq D^*_{x'x}$.

$$\begin{aligned} D_{xx'} &= \delta'(x - x') \\ D^*_{x'x} &= \delta'(x' - x)^* = -\delta'(x - x') \end{aligned}$$

Instead, we can make D Hermitian using a constant complex term: $K = -iD$. Then

$$\begin{aligned} K_{xx'} &= -i\delta'(x - x') \\ K^*_{x'x} &= \left[-i\delta'(x' - x)\right]^* = i\delta'(x' - x) = -i\delta(x - x') \end{aligned}$$

Warning: For operators in infinite dimensions, $K_{xx'} = K^*_{x'x}$ doesn't actually guarantee Hermiticity. To see this, note that a Hermitian operator must satisfy (for functions f, g)

$$\begin{aligned} \langle g|K|f\rangle &= \langle f|K|g\rangle^* \\ \int_a^b \int_a^b \langle g|x\rangle \langle x|K|x'\rangle \langle x'|f\rangle \, dx dx' &\stackrel{?}{=} \left(\int_a^b \int_a^b \langle f|x\rangle \langle x|K|x'\rangle \langle x'|g\rangle \, dx dx' \right)^* \\ \int_a^b g^*(x) \left[-i \frac{df(x)}{dx} \right] dx &\stackrel{?}{=} \left(\int_a^b f^*(x) \left[-i \frac{dg(x)}{dx} \right] dx \right)^* \\ -ig^*(x)f(x)\Big|_a^b + i \int_a^b \frac{dg^*(x)}{dx} f(x) \, dx &\neq i \int_a^b \frac{dg^*(x)}{dx} f(x) \, dx \end{aligned}$$

In order to Hermitian, the **surface term** of K must be 0. K is Hermitian in the space of functions where this holds. $K_{xx'} = K^*_{x'x}$ is no longer a sufficient condition.

$$-ig^*(x)f(x)\Big|_a^b = 0$$

Examples where

- String clamped at one end, $f(0) = f(L) = 0$. Derivative is $K = -i\frac{d}{dx}$
- Periodic function, $f(\phi) = f(\phi + 2\pi)$. Derivative is $K = -i\frac{d}{d\phi}$

In QM, we study functions $-\infty < x < \infty$. There are two classes of such functions.

1. f vanishes as $x \to -\infty$ and $x \to \infty$. Here, $K = -i\frac{d}{dx}$ is Hermitian.
2. f doesn't vanish, like e^{ikx}. If K is Hermitian, it must satisfy

$$e^{ikx}e^{ik'x}\Big|_{-\infty}^{\infty} = 0$$

If $k = k'$, this holds. Otherwise, we define the limit as $|x| \to \infty$ to be the average over a large interval. Then if $k \neq k'$,

$$\lim_{x\to\infty} e^{ikx}e^{ik'x} = \lim_{L,\Delta\to\infty} \frac{1}{\Delta} \int_L^{L+\Delta} e^{i(k-k')x} \, dx = 0$$

Thus K is also Hermitian for class 2 functions.

1.2.2 Eigenvalues of K

For eigenvalue k,

$$\begin{aligned} K\left|k\right\rangle &= k\left|k\right\rangle \\ \left\langle x\right|K\left|k\right\rangle &= k\left\langle x|k\right\rangle \\ \int \left\langle x|K|x'\right\rangle \left\langle x'|k\right\rangle \ dx' &= k\left\langle x|k\right\rangle \end{aligned}$$

Let $\langle x|k\rangle = \psi_k(x)$. Using the definition $K = -iD$,

$$-i\frac{d}{dx}\psi_k(x) = k\psi_k(x)$$

The solution is $\psi_k(x) = Ae^{ikx}$. Thus, any real number k is an eigenvalue of K with eigenvector Ae^{ikx}. Since A is a free parameter, set it to $1/\sqrt{2\pi}$ so that

$$\begin{aligned} \left|k\right\rangle &\leftrightarrow \frac{1}{\sqrt{2\pi}}e^{ikx} \\ \left\langle k|k'\right\rangle = \int_{-\infty}^{\infty} \left\langle k|x\right\rangle \left\langle x|k'\right\rangle dx &= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(k'-k)x}\ dx = \delta(k'-k) \end{aligned}$$

Thus $\langle k|k'\rangle = 0$ if $k \neq k'$, and blows up if $k = k'$.

We require a real eigenvalue k because if k is imaginary, e^{ikx} blows up as $x \to \infty$. The choice of a real eigenvalue restricts us to the **physical Hilbert space**.

1.2.3 Conjugate actions of X and K

In finite dimensions, the eigenfunctions of a Hermitian operator form a complete basis. Assume this holds for infinite dimensions.

K is Hermitian. Expand in both X and K basis. Recall $|k\rangle = \frac{1}{\sqrt{2\pi}}e^{ikx}$.

$$\begin{aligned} f(k) = \left\langle k|f\right\rangle &= \int_{-\infty}^{\infty} \left\langle k|x\right\rangle \left\langle x|f\right\rangle dx = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ikx}f(x)\ dx \\ f(x) = \left\langle x|f\right\rangle &= \int_{-\infty}^{\infty} \left\langle x|k\right\rangle \left\langle k|f\right\rangle dk = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{ikx}f(k)\ dk \end{aligned}$$

The Fourier transform pass f from one complete basis to another.

Additionally, note the matrix elements of X and K in the X and K basis, respectively:

$$\begin{aligned} \left\langle x|X|x'\right\rangle &= x'\delta(x-x') \\ \left\langle k|K|k'\right\rangle &= k'\delta(k-k') \end{aligned}$$

In finite dimensions, we had (in the X basis) $X\left|x\right\rangle = x\left|x\right\rangle$. To see the action of X in infinite dimensions, let $X\left|f\right\rangle = \left|\tilde{f}\right\rangle$.

$$\begin{aligned}
\langle x|X|f\rangle &= \int_{-\infty}^{\infty} \langle x|X|x'\rangle \langle x'|f\rangle \, dx' = xf(x) \\
\langle x|X|f\rangle &= \langle x|\tilde{f}\rangle = \tilde{f}(x)
\end{aligned}$$

Thus $\tilde{f}(x) = xf(x)$, and the effect of X in infinite dimensions is $X\left|f(x)\right\rangle = \left|xf(x)\right\rangle$. That is, multiplying $f(x)$ by x.

An interesting phenomenon occurs if we calculate the matrix elements of X in the K basis. Recall $\left|k\right\rangle = \frac{1}{\sqrt{2\pi}}e^{ikx}$.

$$\begin{aligned}
\langle k|X|k'\rangle &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \langle k|x\rangle \langle x|X|x'\rangle \langle x'|k'\rangle \, dxdx' \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x'\delta(x-x') \langle k|x\rangle \langle x'|k'\rangle \, dxdx' \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ikx}e^{ik'x} \, dx \\
&= i\frac{d}{dk}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(k'-k)x} \, dx\right) = i\delta'(k-k')
\end{aligned}$$

Thus, if g is a function with image $g(k)$ in the K basis,

$$X\left|g(k)\right\rangle = \left|i\frac{dg(k)}{dk}\right\rangle$$

To summarize,

$$\text{In } X \text{ basis } \begin{cases} X \equiv x \\ K \equiv -i\frac{d}{dx} \end{cases}$$

$$\text{In } K \text{ basis } \begin{cases} X \equiv i\frac{d}{dx} \\ K \equiv k \end{cases}$$

Operators like these are **conjugate** to each other.

Note that X and K do not commute.

$$\begin{aligned}
XK\left|f\right\rangle &= -ix\frac{df(x)}{dx} \\
KX\left|f\right\rangle &= -i\frac{d}{dx}\left(xf(x)\right) = -ix\frac{df(x)}{dx} - if(x)
\end{aligned}$$

Thus $[X,K]\left|f\right\rangle = if = iI\left|f\right\rangle$. Since f was arbitrary, we have $[X,K] = iI$.

In QM, X and K are the main operators we'll be seeing (X gives position, K gives derivative). However, they take different forms: $\hat{X} = X$ and $\hat{P} = \hbar K$.

2 Review of Classical Mechanics (pp. 75 – 106)

2.1 Lagrangian Mechanics

Newtonian mechanics can find the path of a particle at an infinitesimal time slice:

$$m\ddot{x} = -\frac{dV}{dx}$$

Given $x(t_i)$ and $\dot{x}(t_i)$ at a point t_i, we can obtain the classical path's $x_{\text{cl}}(t_i + \Delta t)$ and $\dot{x}_{\text{cl}}(t_i + \Delta t)$ after some infinitesimal time Δt using the equation above. By continuing this process, we recreate the classical path x_{cl} of the particle.

$$x_{\text{cl}}(t_i + \Delta t) = x(t_i) + \dot{x}(t_i)\Delta t$$
$$\dot{x}_{\text{cl}}(t_i + \Delta t) = \dot{x}(t_i) + \ddot{x}(t_i)\Delta t$$

Lagrangian mechanics seeks to find the classical path immediately. Given two points (x_i, t_i) and (x_f, t_f), we can find the path taken by the particle between them. But what distinguishes the classical path from all other possible ones?

2.1.1 Procedure

1. Define the Lagrangian $L = T - V$. Assume time-independent potential, thus $L = L(x, \dot{x})$.
2. For each path from (x_i, t_i) to (x_f, t_f), define the action as a function of the entire path:

$$S[x(t)] = \int_{t_i}^{t_f} L(x, \dot{x})\, dt$$

3. **Principle of least action:** the classical path is that on which S is minimized.

NB: assumption of a time-independent potential doesn't always hold. For example, $F_B = qv \times B$. So $L = T - V$ isn't always the correct form.

2.1.2 The Euler-Lagrange Equation

For finite variables, the function was at a extremal if the 1st order change δf vanishes:

$$f(\mathbf{x^0} + \eta) = f(\mathbf{x^0}) + \sum_{i=1}^{n} \left.\frac{\partial f}{\partial x_i}\right|_{\mathbf{x^0}} \eta_i + O(\eta^2)$$
$$\delta f = \sum_{i=1}^{n} \left.\frac{\partial f}{\partial x_i}\right|_{\mathbf{x^0}} \eta_i = 0$$

Assume this holds for functions of infinite variables, like $S[x(t)]$. Let $x_{\text{cl}}(t)$ be the classical path, and $x_{\text{cl}}(t) + \eta(t)$ a nearby path. Endpoints must math, so $\eta(t_i) = \eta(t_f) = 0$. Now find the extremal where

1st order change δS vanishes:

$$\begin{aligned}
S[x_{\text{cl}}(t)+\eta(t)] &= \int_{t_i}^{t_f} L(x_{\text{cl}}(t)+\eta(t), \dot{x}_{\text{cl}}(t)+\dot{\eta}(t))\, dt \\
&= \int_{t_i}^{t_f} \left[L(x_{\text{cl}}(t), \dot{x}_{\text{cl}}(t)) + \left.\frac{\partial L}{\partial x(t)}\right|_{x_{\text{cl}}} \cdot \eta(t) + \left.\frac{\partial L}{\partial \dot{x}(t)}\right|_{x_{\text{cl}}} \cdot \dot{\eta}(t) + \dots \right] dt \\
&\approx S[x_{\text{cl}}(t)] + \delta S
\end{aligned}$$

Find where $\delta S = 0$ for an arbitrary deviation η:

$$\begin{aligned}
0 = \delta S &= \int_{t_i}^{t_f} \left[\left.\frac{\partial L}{\partial x(t)}\right|_{x_{\text{cl}}} \cdot \eta(t) + \left.\frac{\partial L}{\partial \dot{x}(t)}\right|_{x_{\text{cl}}} \cdot \dot{\eta}(t) \right] dt \\
&= \int_{t_i}^{t_f} \left[\left.\frac{\partial L}{\partial x(t)}\right|_{x_{\text{cl}}} \cdot \eta(t) \right] dt + \left[\left.\frac{\partial L}{\partial \dot{x}(t)}\right|_{x_{\text{cl}}} \cdot \eta(t) \right]_{t_i}^{t_f} - \int_{t_i}^{t_f} \left[\frac{d}{dt} \left.\frac{\partial L}{\partial \dot{x}(t)}\right|_{x_{\text{cl}}} \cdot \eta(t) \right] dt \\
&= \int_{t_i}^{t_f} \left[\frac{\partial L}{\partial x(t)} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}(t)} \right]_{x_{\text{cl}}} \cdot \eta(t)\, dt
\end{aligned}$$

Thus we get the Euler-Lagrange equation, which must be satisfied by a classical path:

$$\boxed{\left[\frac{\partial L}{\partial x(t)} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}(t)} \right]_{x_{\text{cl}}} = 0}$$

2.1.3 Conservation Laws from the Lagrangian

A benefit of the Lagrangian is that unlike the Newtonian, coordinates don't have to be Cartesian. We can have an arbitrary q_i and $\dot{q}_i$.

Further note:

- Canonical momentum conjugate: $p_i = \partial L / \partial \dot{q}_i$
- Canonical force conjugate: $F_i = \partial L / \partial q_i$

$$F_i = \frac{d}{dt} p_i$$

E.g. if $q_i = \theta$, then p_i is angular momentum and F_i is torque.

Call q_i a **cyclic coordinate** if L depends on $\dot{q}_i$, but not q_i. Then p_i is conserved:

$$\frac{dp_i}{dt} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i} = 0$$

Conservation laws fall out naturally out of the Lagrangian, while it may be opaque in the Newtonian. For example, consider a Lagrangian in polar coordinates $L = L(\rho, \phi)$. The metric is $ds^2 = d\rho^2 + \rho d\phi^2$. Thus the infinitesimal velocity is

$$v = \frac{ds}{dt} = \sqrt{(\dot{\rho})^2 + (\rho\dot{\phi})^2}$$

The Lagrangian is

$$L = \frac{1}{2}m\left[\dot{\rho}^2 + \rho^2\dot{\phi}^2\right] - V(\rho,\phi)$$

Applying Euler-Lagrange for ρ and ϕ respectively gives

$$\begin{aligned}
\frac{d}{dt}(m\dot{\rho}) = -\frac{\partial V}{\partial \rho} + m\rho\dot{\phi}^2 \quad &\longrightarrow \quad m\ddot{\rho} = -\frac{\partial V}{\partial \rho} + m\rho(\dot{\phi})^2 \\
\frac{d}{dt}\left(m\rho^2\dot{\phi}\right) = -\frac{\partial V}{\partial \phi} \quad &\longrightarrow \quad m\ddot{\phi} = -\frac{1}{\rho^2}\frac{\partial V}{\partial \phi} - \frac{1}{\rho}(2m\dot{\rho}\dot{\phi})
\end{aligned}$$

Note how easily we can derive the centrifugal and coriolis forces. Also note that conservation laws for cyclic coordinates $V(\rho,\phi) = V(\rho)$ and $V(\rho,\phi) = V(\phi)$ fall out naturally, and the corresponding equations of motion are that of $\partial V/\partial\phi = 0$ and $\partial V/\partial\rho = 0$.

In sum:

- Equations of motion are simpler to find with Lagrangian
- Conservation laws are easy to see from Lagrangian
- Path integral formalism uses Lagrangian (later)

2.1.4 The Electromagnetic Lagrangian

Electromagnetic force:

$$\vec{F} = q\left(\vec{E} + \frac{\vec{v}}{c}\times\vec{B}\right)$$

Note: this is Gaussian units, hence the $1/c$ factor for magnetic force.

The electromagnetic force is velocity-dependent, so $L = T - V$ doesn't work. Instead, we construct L to reduce to the expected equations. The EM lagrangian is

$$L_{\text{EM}} = \frac{1}{2}m\vec{v}\cdot\vec{v} - q\phi + \frac{q}{c}\vec{v}\cdot\vec{A}$$

ϕ and $\vec{A}$ are the scalar and vector potentials, related to E and B by

$$\begin{aligned}
\vec{E} &= -\nabla\phi - \frac{1}{c}\frac{\partial\vec{A}}{\partial t} \\
\vec{B} &= \nabla\times\vec{A}
\end{aligned}$$

The Euler-Lagrange equation is

$$\begin{aligned}
\frac{d}{dt}\left(m\dot{x}_i + \frac{q}{c}A_i\right) &= -q\frac{\partial\phi}{\partial x_i} + \frac{q}{c}\frac{\partial(\vec{v}\cdot\vec{A})}{\partial x_i}\ , \quad i = 1,2,3 \\
\frac{d}{dt}\left(m\vec{v} + \frac{q}{c}\vec{A}\right) &= -q\nabla\phi + \frac{q}{c}\nabla(\vec{v}\cdot\vec{A}) \\
\frac{d}{dt}(m\vec{v}) &= -q\nabla\phi + \frac{q}{c}\left[-\frac{d\vec{A}}{dt} + \nabla(\vec{v}\cdot\vec{A})\right] \qquad (1)
\end{aligned}$$

By time-dependence of the magnetic vector potential $\vec{A}$,

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + \sum_i \frac{\partial \vec{A}}{\partial x_i}\frac{dx_i}{dt} = \frac{\partial \vec{A}}{\partial t} + (\vec{v}\cdot\nabla)\vec{A}$$

Thus equation (1) becomes

$$\frac{d}{dt}(m\vec{v}) = -q\nabla\phi - \frac{q}{c}\frac{\partial \vec{A}}{\partial t} + \frac{q}{c}\underbrace{\left[\nabla(\vec{v}\cdot\vec{A}) - (\vec{v}\cdot\nabla)\vec{A}\right]}_{\vec{v}\times(\nabla\times\vec{A})} \tag{2}$$

This reduces to $\vec{F} = q\vec{E} + \frac{q}{c}\vec{v}\times\vec{B}$.

The time-dependent EM field is usually not conservative, so there's no path-independent work function to use as the potential. Thus L_{EM} is made to fit the situation and reduce as expected. $L = T - V$ only works for conservative forces.

2.1.5 The Two-Body Problem

Assume 2 masses m_1, m_2 that exert equal and opposite forces on each other. We can reduce this to a one-body problem with Lagrangian mechanics.

The potential only depends on relative position, $V(\vec{r}_1, \vec{r}_2) = V(\vec{r}_1 - \vec{r}_2)$. Thus $6 \to 3$ cyclic coordinates in 3D.

The two-body Lagrangian is

$$L = \frac{1}{2}m_1|\dot{r}_1|^2 + \frac{1}{2}m_2|\dot{r}_2|^2 - V(r_1 - r_2) \tag{1}$$

Define $\vec{r}_{\text{CM}}$ and $\vec{r}$ such that

$$\vec{r}_{\text{CM}} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}\ ,\quad \vec{r} = \vec{r}_1 - \vec{r}_2$$
$$\vec{r}_1 = \vec{r}_{\text{CM}} + \frac{m_2\vec{r}}{m_1 + m_2}$$
$$\vec{r}_2 = \vec{r}_{\text{CM}} - \frac{m_1\vec{r}}{m_1 + m_2}$$

The two-body Lagrangian becomes

$$L = \frac{1}{2}(m_1 + m_2)|\dot{r}_{\text{CM}}|^2 + \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}|\dot{r}|^2 - V(\vec{r}) \tag{2}$$

This can be thought of as the Lagrangian of two fictious particles that don't interact: one "CM" particle with mass $M = m_1 + m_2$, and one "reduced mass" particle with mass $\mu = (m_1 m_2)/(m_1 + m_2)$. Note that $L(r, \dot{r}, r_{\text{CM}}, \dot{r}_{\text{CM}}) = L(r, \dot{r}) + L(r_{\text{CM}}, \dot{r}_{\text{CM}})$.

By going to the CM frame, we can eliminate $\dot{r}_{\text{CM}}$ and just solve for the motion of μ. We've reduced the two-body problem to a one-body problem.